SREEKAR PRANEETH MARRI
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EDUCATION

Scuola Superiore Sant'Anna — Pisa, Italy
PhD in BioRobotics — Oct 2026 - Present

Boston University — Boston, Massachusetts
MS (Thesis) in Robotics & Autonomous Systems with Specialization in Data Analytics; GPA: 3.88/4.0 — Aug 2024 - May 2026
Thesis: *"Real-Time Control of Multi-DOF Soft Robotic Actuators using Machine Learning and Model Predictive Control"*

University of California, Riverside — Riverside, California
Advanced Certificate in Engineering (Computer Science, Academic Transfer); GPA: 3.96/4.0 — Sep 2023 - Jun 2024

Amrita School of Engineering, Bangalore — Bengaluru, India
BTech in Computer Science Engineering (Artificial Intelligence) with Distinction; GPA: 9.15/10.0 — Sep 2020 - Aug 2023

PROFESSIONAL EXPERIENCE

Brain Inspired Robotics Lab, Institute of BioRobotics — Pisa, Italy
Doctoral Researcher for Prof. Egidio Falotico. — Oct 2026 - Present

- Developing vision-language based controls for soft continuum robot inspired by elephant trunk.

Morphable Biorobotics Lab, Boston University — Boston, Massachusetts
Graduate Researcher for Prof. Tommaso Ranzani — Sep 2024 - Aug 2026

- Developing inverse models and reinforcement learning frameworks for precise open-loop and closed-loop control of multi-DoF soft actuators in underwater environments.
- Designing a real-time control architecture for multi-chamber Stacked Balloon Actuators (SBAs) with integrated sensing and combining reinforcement learning for adaptive motion control.
- Advancing data-driven modelling and trajectory-tracking methods to improve performance of soft robotic systems.

Tay Lab, Boston University School of Medicine — Boston, Massachusetts
Research Programmer for Prof. Tuan Leng Tay — Nov 2024 - Mar 2025

- Applied deep learning to high-resolution microscopy data to automate analysis of microglial cellular structures.
- Designed and implemented and end-to-end U-Net segmentation pipeline for robust microglial cell detection and feature extraction, enabling scalable, high-throughput image processing.
- Managed the full MLOps workflow - from large-scale annotation to preprocessing to experimental tracking and deploying on HPC infrastructure, ensuring reproducibility and production-grade model training and evaluation.

Indian Institute of Science (IISc), Bangalore — Bengaluru, India
AI Research Intern under Prof. Punit Rathore — Jun 2023 - Sep 2023

- Fine-tuned a LLaMA2-based legal document assistant by developing advanced natural language processing (NLP) pipelines and creating robust web-scraping workflows to collect and curate high-quality, domain-specific legal datasets.
- Evaluated and benchmarked large language model (LLM) outputs for accuracy, contextual relevance, and coverage of domain-specific legal knowledge to ensure reliable, high-performing results.
- Conducted a comprehensive systematic literature review on road condition classification using smartphone sensor data, employing signal processing to extract relevant features and enhance classification accuracy.

New Space Research Technologies — Bengaluru, India
Robotic Intern — Jul 2022 - Sep 2022

- Simulated and validated quadcopter operations on real UAV hardware using the Robot Operating System (ROS) and Software-in-the-Loop (SITL) environments.
- Implemented and optimized flight planning and control algorithms to improve mission stability and precise path tracking.
- Integrated ROS-based Python modules for autonomous drone control, ensuring reliable communication between SITL simulations and physical quadcopter systems.

PUBLICATIONS

- H.S. Chilakala, N. Preeti, **Sreekar Praneeth Marri**, K. Murali, "Kalman Integration with ARIMA & LSTM," *Advances in Networks, Intelligence and Computing*, 1st Edition, CRC Press, 2024, 8 pages, eBook ISBN 9781003430421, DOI: 10.1201/9781003430421-57.
- H.S. Chilakala, N. Preeti, **S.P. Marri**, S. Palaniswamy, P.B. Pati, "Real-Time Object Detection Using Mobile Robot Captured Images: A Deep Learning Approach," *2024 3rd International Conference for Innovation in Technology (INOCON)*, Bangalore, India, 2024, pp. 1-6, DOI: 10.1109/INOCON60754.2024.10511303.

- **S.P. Marri**, B.V. Nikith, N.K.S. Keerthan, S. Jayan, "SVM Versus KNN: Prediction of Best Image Classifier," *In: Reddy V.S., Prasad V.K., Wang J., Rao Dasari N.M. (eds) Intelligent Systems and Sustainable Computing. ICISSC 2022*. Smart Innovation, Systems and Technologies, vol. 363, Springer, Singapore, 2023, DOI: 10.1007/978-981-99-4717-1_14.
- B.V. Nikith, N.K.S. Keerthan, **M.S. Praneeth**, Dr. T. Amrita, "Leaf Disease Detection and Classification," *Procedia Computer Science*, vol. 218, pp. 291-300, 2023, ISSN 1877-0509, DOI: 10.1016/j.procs.2023.01.011.
- N.K.S. Keerthan, **S.P. Marri**, M. Khanna, "Analysis of Key Based Cryptographic Algorithms and Its Applications," *2023 IEEE 3rd International Conference on Technology, Engineering, Management for Societal Impact using Marketing, Entrepreneurship and Talent (TEMSMET)*, Mysuru, India, 2023, pp. 1-4, DOI: 10.1109/TEMSMET56707.2023.10150061.

PROJECT EXPERIENCE

Edge-Optimized LLM for Autonomous Robotics (Llama3, Whisper, ROS)
- Integrated quantized Llama3 and Whisper models onto a mobile robotic platform to enable speech-driven, offline navigation with ROS, eliminating cloud dependencies for real-time performance.
- Demonstrated 96% positional and 93% rotational command accuracy at just 400ms inference latency, setting a benchmark for embedded language models in robotics.

Reinforcement Learning-Based Robotic Arm for Color Sorting (ROS, Python)
- Led a team of three to develop a 5-DOF robotic arm for autonomous color-based object sorting using reinforcement learning with adaptive reward strategies.
- Linked simulation and hardware deployments to iteratively refine control policies, reducing misclassification and improving sorting consistency.

LIGO Time-Series Anomaly Detection with Matrix Profiles (Python, Signal Processing)
- Applied matrix profiling techniques to LIGO gravitational datasets, detecting anomalies linked to astrophysical events.
- Built statistical workflows to improve event detection and cosmic classification accuracy.

Real-Time Flight Delay Prediction System (Spark, Kafka, MongoDB, Hadoop)
- Designed a scalable big-data pipeline for live flight delay prediction using Spark Structured Streaming and Kafka.
- Improved prediction accuracy by 20% over baselines, enabling better scheduling and resource allocation.

SKILLS

Robotics & AI: ROS, MATLAB, Simulink, SITL, PyTorch, TensorFlow, Transformers, OpenAI Gym, LLaMA, BERT Models
Programming & ML : Python, SQL, Statistical Modelling, Data Pipelines, Regression, Matrix Profiles, KPI Analysis
Prototyping & Systems: 3D Printing, Soft Actuator Fabrication, AutoCAD, Fusion 360, Adobe Illustrator
Big Data Tools: Spark, Kafka, Hadoop, MongoDB

AWARDS & ACHIEVEMENTS

Dean's Scholarship Recipient, Boston University College of Engineering — 2024 - 2026
Ranked **7th in Programme**, Amrita School of Engineering, Bengaluru — 2024
Placed in the **Top 25** of the **Honeywell Smart City Hackathon**, Bengaluru — 2021

TEACHNING EXPERIENCE

Graduate Teaching Assistant, *EK 131 - Introduction to Engineering,* Boston University — Spring 2026
Graduate Teaching Assistant, *ME 568 - Soft Robotic Technologies*, Boston University — Fall 2025
Graduate Course Grader*, CS 541, CS 542, CS 332,* Boston University

POSITION OF RESPONSIBILITY

Student Ambassador, *Robotics and Autonomous Systems*, Boston University — Jan 2025 - May 2026
Joint Secretary, *FACE (CS) club*, Amrita School of Engineering — Sep 2021 - Aug 2022
Class Representative, *CSE (AI)*, Amrita School of Engineering — Sep 2020 - Aug 2023

COMMUNITY SERVICE

Live-In-Labs Program
Rural Development and Sustainable Initiative — Jan 2023
- Conducted field surveys and data analysis in a rural village to identify socio-economic and infrastructural challenges.
- Designed and implemented sustainable, low-cost solutions to improve community living standards.

CERTIFICATION

Indian Institute of Technology (IIT), Madras
Online B.Sc. in Programming and Data Science, Completed Foundational Level — Jan 2023